

SECTION 13855-A
VERY EARLY WARNING SMOKE DETECTION SYSTEM

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. Furnish and install a complete and operational very early warning smoke detection (VEWSD).
- B. The system shall include all wiring, raceways, pull boxes, terminal cabinets, outlet and mounting boxes, control equipment, smoke detectors, sampling pipe network, and all other accessories and miscellaneous items required for a complete system, even though each item is not specifically mentioned or indicated.
- C. Interface and connection to the building fire alarm and smoke detection system are included.
- D. Design piping configuration and layout.

1.2 RELATED WORK

- A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for EWSD:
 - 2. Section 13851 - Fire Alarm and Smoke Detection Systems.
 - 3. Section 16010 - General Electrical Provisions.
- B. CAUTION: Use of this Section without including all of the above-listed items will result in omission of basic requirements.
- C. In the event of conflict regarding VEWSD requirements between this Section and any other section, the provisions of this Section shall govern.

1.3 SYSTEM DESCRIPTION

- A. The VEWSD system shall interface with the building fire alarm system, Section 13851.
- B. The smoke detection system shall operate in a local automatic fashion.

- C. Contractor to ensure that the software and hardware to be installed shall be capable with data functionality as year-2013 compliant.
- D. The system shall provide very early warning of a fire in its incipient stage through sampling of air from the protected area.
- E. Sampling of air shall be continuous for detector sensing. Detector output shall be an analog signal corresponding to the concentration of smoke in the sampled air.
- F. The smoke detector assembly output signal shall be transmitted to the appropriate VEWS control panel with display annunciator. The annunciator shall provide visual indication of the smoke concentration in the sampled air.
- G. The VEWS control panel shall contain, but not be limited to, a control card for each detector it supports. The control cards shall include three field-programmable alarm levels with one Form C relay contact for each level.

1.4 OPERATION AND MAINTENANCE DATA

- A. Provide as-installed schematic, wiring, and point-to-point interconnection diagrams of all circuits, internal and external, for all equipment installed. These schematics shall include the conductor color-coding system. Drawings shall be AutoCAD, 2007.
- B. Provide complete, as-installed riser diagrams indicating the wiring sequence of all alarm-initiating devices. Drawings shall be AutoCAD, 2007.
- C. Provide a complete description of the system operation, including a schedule of relay abbreviations used on the Drawings, list of relay functions, and the sequence of relay operation during supervisory, trouble, and alarm conditions.
- D. Provide final calculations and completed system checkout field report with copies of system response tests.
- E. Include operating instructions and maintenance and repair procedures.
- F. Include manufacturer's letter stating that system is operational.

- G. Provide manufacturer's certificate that system meets or exceeds specified requirements.
 - H. Provide control panel internal wiring diagram with function description for incoming and outgoing terminal.
- 1.5 EXTRA MATERIALS
- A. Provide two keys of each type (when applicable).
- 1.6 SUBMITTALS
- A. Prior to installation, provide engineering calculations that include the maximum smoke transport time from the last sampling point and airflow balance details from each sampling point.
 - B. Provide installation drawings showing details of the sampling pipe network proposed layout, piping isometrics, and sampling hole sizes and locations.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Xtralis.
- B. System contractor and existing shall have the same manufactures/product to integrate both portions as a completed system.

2.2 SAMPLING PIPE

- A. Length of pipe and sampling hole spacings shall be sized for required areas of coverage. Sample piping length, diameter, material of construction, sample hole size, and location shall be in accordance with manufacturer's installation instructions.
- B. The sampling pipe shall be constructed of Schedule 40 CPVC pipe with glued joints or EMT conduit as specified on the construction drawings. Other types of sampling pipe material shall be considered for special applications subject to the approval of the Engineer and/or Owner.
- C. Maximum time to activate a high alarm from canned smoke

exposed 1 meter from the furthest sample point shall be 120 seconds.

2.3 DETECTOR ASSEMBLY

- A. The following are minimum requirements:
 - 1. Power Supply: 120 Vac, single phase, 60 Hz.
 - 2. Operating Voltage: 24 Vdc (20 Vdc minimum, 28 Vdc maximum).
 - 3. Sensitivity: 0.003 through 0.15 percent obscuration per foot.
 - 4. Ambient Environment: 32 degrees F to 125 degrees F.
 - 5. Relative Humidity: zero percent to 95 percent noncondensing.
 - 6. Coverage: 20,000 square feet.
 - 7. Approvals/Listing:
 - a. Factory Mutual (FM) approved.
 - b. Underwriter's Laboratories (UL) listed.

2.4 CONTROL PANEL

- A. The following are minimum requirements:
 - 1. Power Supply: 120 Vac, single phase, 60 Hz.
 - 2. Operating Voltage: 24 Vdc (20 Vdc minimum, 28 Vdc maximum).
 - 3. Ambient Environment: 32 degrees F to 125 degrees F.
 - 4. Relative Humidity: zero percent to 95 percent noncondensing.
 - 5. Alarm and Trouble Contacts: three programmable Form C alarm contacts, zero- to 60-second programmable time delay, available in latched or nonlatched contacts; one Form C trouble contact to indicate detector status, airflow status, primary power and battery status, isolate/active status; available in latched or nonlatched contacts.
 - 6. Panel Control/Indication: alarm and trouble status indication; bar graph; sensitivity test switch; normal/isolation (test) switch.
 - 7. Approvals/Listing:
 - a. FM approved.
 - b. UL listed.

2.5 POWER SUPPLY

- A. Adequate to serve control panel and remote detectors, including battery-operated emergency power supply with capacity for operating system in standby mode (including a 20 percent safety factor) for 4 hours (in conjunction with an automatic-starting engine-driven generator) followed by alarm mode for 10 minutes. Batteries shall be

sealed, maintenance free type. Loss of normal power shall be reported as a trouble to fire alarm panel.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Install equipment and wiring in accordance with the schedule on the Drawings and the manufacturer's diagrams and recommendations.
- B. Connections for all external wiring between the control unit and all external devices shall be made on terminal strips installed in accessible locked terminal cabinets. Connections may be made either with terminal spade lugs set on the conductors with a special setting tool or with approved pressure-type terminal blocks.
- C. All external wiring shall be color coded and shall not be installed in the same outlet box, junction box, or conduit with conductors of lighting or power systems. Splicing shall not be permitted.
- D. Locate and install the detector assembly for optimum response time.
- E. Install the sampling pipe network in accordance with the manufacturer's recommendations. Make all necessary adjustments for optimal air sampling.
- F. Layout of the sampling pipe network shall provide for a cross-sectional sampling of the return airstream. Careful consideration shall be given to NFPA 72 guidelines to the number of sampling points and spacing with regard to high ceilings and forced-air ventilation applications.
- G. Coordinate the mounting and penetration of sampling pipe with mechanical equipment installation and structural elements.
- H. The sampling pipe network shall be cleaned and swabbed of all debris prior to connection and operation of detector assembly.
- I. Exposed sampling pipe areas shall have the exterior cleaned of all dirt and markings.

- J. Label sampling pipe with adhesive labels from manufacturer every 10 feet maximum. Identify the location of sampling orifices with a visible marking.
- K. Sampling pipe joints shall be airtight and adequately supported. Joint sealers shall be suitable for the area of installation per Section 07900, Joint Sealers.

3.2 QUALITY CONTROL

- A. The system shall be designed, installed, tested, and adjusted under the supervision of the manufacturer.
- B. Ten days prior to the test, a data logger shall be installed to monitor background ambient conditions for each zone. Document all tests.
- C. At the completion of the installation, the system shall be tested in accordance with the manufacturer's test and commissioning procedures in the presence of the Owner and under the supervision of the manufacturer. Testing shall include the following:
 - 1. The complete system shall be tested free from grounded or open circuits.
 - 2. Perform airflow, fault, and sensitivity tests on the completed installed system. Perform time-delay tests for detector response using a data logger. Adjust the time-delay settings to avoid unwanted alarms. Document all tests.
 - 3. Perform tests on system response to verify it is within the specified range. Expose smoke to the farthest sampling point and record system response time. Check for sampling pipe integrity and pipe blockage. Correct any problems and repeat test.
 - 4. Perform tests to indicate that all alarm-initiating devices and supervisory devices are operating normally and smoke detection system output signal is tied into the building fire detection and alarm system.
- D. The test and report procedure shall be prepared by the system manufacturer and reviewed by the Engineer.
- E. Any defects shall be corrected at once and the test reconducted at no additional cost to the Owner.

3.3 MANUFACTURER'S FIELD SERVICES

- A. Include services of factory-certified technician to supervise installation, adjustments, final connections,

and system testing.

- B. The manufacturer shall provide the Owner instruction in the operation and maintenance of the system and test equipment.

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